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Learning Seminar on Modular Forms, Automorphic Representations, and L-functions

Monday Sep 17, 2026

§2.4 "Basic" Representation Theory of Automorphic Forms and Representations

a Lie group

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Def: A representation (π, H) of G is an ordered pair in which

- H is a topological vector space and
- $\pi: G \rightarrow \text{End}(H)$ is a homomorphism such that the map
 $G \times H \rightarrow H$ given by
 $(g, f) \mapsto \pi(g)f$ is continuous.

Note: Often abbreviate (π, H) as π or H

Def: If H is a Hilbert space, the representation is called unitary if, furthermore, $\pi(g): H \rightarrow H$ is a unitary operator for all $g \in G$.

Recall: A Hilbert space is a real or complex inner product space that is also a complete metric space with respect to the distance function induced by the inner product.

→ Note: All Hilbert spaces will be assumed complex.

- Write $\langle -, - \rangle$ to denote the given Hermitian inner product on H .
- Then π is unitary if $\langle f, f' \rangle = \langle \pi(g)f, \pi(g)f' \rangle$
 or equivalently $\langle f, f \rangle = \langle \pi(g)f, \pi(g)f \rangle$ for all $f, f' \in H, g \in G$
- Write $\|f\| = \sqrt{\langle f, f \rangle}$ to denote the norm (real valued fn).

Fundamental Motivating Problem in Representation Theory:

Given a Lie group G , classify all unitary representations of G .

- The set of all unitary representations of G , up to isomorphism, is referred to as the unitary dual of G .
- Unitary Dual Problem

Our Goal in the next 4 seminar talks: Compute the unitary dual of
 $G = \text{SL}(2, \mathbb{R}), \text{GL}(2, \mathbb{R})^+, \text{GL}(3, \mathbb{R})$

why? Relevant for number theory.

→ The first person to compute the unitary dual of $G = \text{SL}(2, \mathbb{R})$ was a theoretical physicist Valentin Bargmann

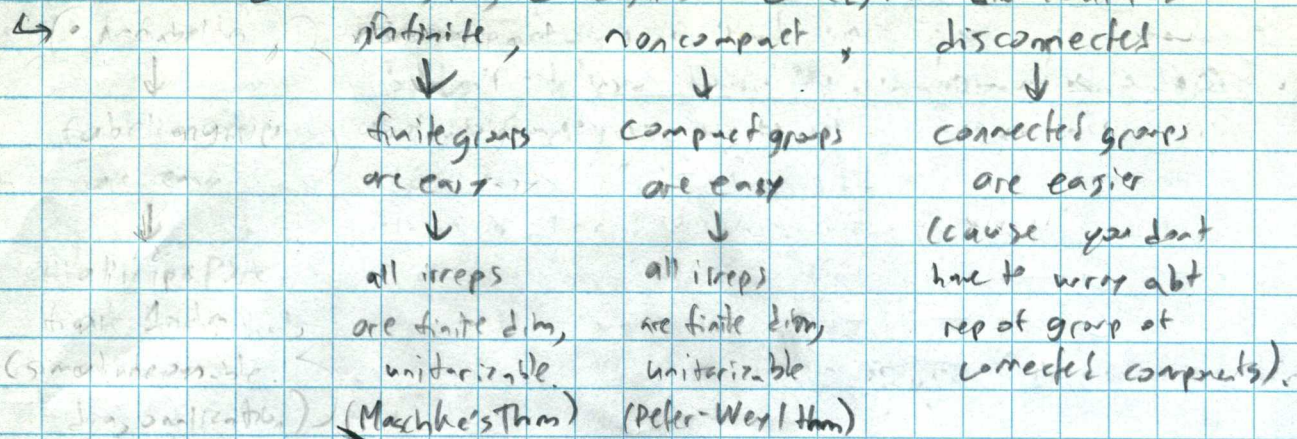
- "Irreducible unitary representations of the Lorentz group" (1947)
 Annals of Mathematics.

→ We will follow Harish-Chandra's approach instead.

Today: Develop the standard notions of smooth vectors, K -finite vectors, admissible representations, $(\mathfrak{g}, K)$ -modules, infinitesimal equivalence

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what makes $G = SL(2, \mathbb{R}), GL(2, \mathbb{R})^+, GL(2, \mathbb{R})$ difficult?



Thm 2.4.1 (Peter-Weyl): Let K be a compact group. (subgroup of $GL(n, \mathbb{C})$)

- (i) The matrix coefficients of finite dimensional unitary representations of K are dense in $C(K)$ and in $L^p(K)$ for all $1 \leq p < \infty$.
- (ii) Any irreducible unitary representation of K is finite dimensional.
- (iii) Let (π, H) be a unitary representation of K . Then H decomposes into a (Hilbert space) direct sum of irreducible unitary subrepresentations.

- If (π, H) is a representation of G on a Hilbert space, a matrix coefficient of the representation is a function of the form $g \mapsto \langle \pi(g)x, y \rangle, x, y \in H$.
- $C(K)$ is the ring of continuous functions on K , given the L^∞ norm.

Lemma 2.4.3: Let (π, H) be a representation of a compact group K on a Hilbert space. There exists a Hermitian inner product $\langle \cdot, \cdot \rangle$ on H inducing the same topology as the original one and satisfying $\langle \pi(k)v, \pi(k)w \rangle = \langle v, w \rangle$ for all $k \in K$.

Proposition 2.4.3:

Let G be a compact group, and let $(\pi_1, H_1), (\pi_2, H_2)$ be representations. Assume that (π_2, H_2) is unitary. If there exists matrix coefficients f_1, f_2 of π_1, π_2 that are not orthogonal in $L^2(G)$, then there exists a nonzero intertwining operator $L: H_1 \rightarrow H_2$.

- If $(\pi_1, H_1), (\pi_2, H_2)$ are representations of G , an intertwining operator is a continuous linear map $L: H_1 \rightarrow H_2$ such that $\pi_2(g) \circ L = L \circ \pi_1(g)$ for all $g \in G$.

- $\hookrightarrow$ homomorphism of G -representations.
- $\hookrightarrow$ aka. G -equivariant map.

- Not covered in Bump: Cartan-Weyl theory: Classify all (finite dim) irrep of K by classifying Lie algebra representations of $\mathfrak{h}_0 = \text{Lie}(K)$ or its complexification $\mathfrak{h}$ (semisimple complex Lie algebra).

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- Cartan - Weyl theory allows you to classify irreducible representations of complex semisimple Lie algebras using the Theorem of Cartan & Weyl.
- Weyl's Unitarian Trick allows you to reduce the classification of finite dimensional irreducible representations of a ^{connected} semisimple (or reductive) Lie group G (ex. $G = SL(2, \mathbb{R})$, $GL(2, \mathbb{R})^+$) to the compact case.
- Note: None of the nontrivial finite dimensional irreps of semisimple G (ex $G = SL(2, \mathbb{R})$) are unitary.

Harish-Chandra's approach:

- Use the Lie algebra $\mathfrak{g} = \mathfrak{g}_0 \oplus_{\mathbb{R}} \mathbb{C}$, $\mathfrak{g}_0 = \text{Lie}(G)$ and the maximal compact $K \leq G$ to study infinite dimensional representations of G .
- Ex: when $G = SL(2, \mathbb{R})$, $\mathfrak{g}_0 = \mathfrak{sl}(2, \mathbb{R})$, $K = SO(2)$.
- $G = GL(2, \mathbb{R})^+$, $\mathfrak{g}_0 = \mathfrak{gl}(2, \mathbb{R})$, $K = SO(2)$
- $G = GL(2, \mathbb{R})$, $\mathfrak{g}_0 = \mathfrak{gl}(2, \mathbb{R})$, $K = O(2)$.

Technical

Obstacle: When H is infinite dimensional, can't differentiate the Lie group action $G \curvearrowright H$ to get a Lie algebra representation $\mathfrak{g} \curvearrowright H$. $\triangle X$

Technical Resolution: Define a dense linear subspace of H , the space H^∞ of smooth vectors.

Definition: We say a vector $f \in H$ is C^1 if for all $X \in \mathfrak{g}$,

$$\pi(X)f = Xf = \frac{d}{dt} \pi(\exp(tX))f \Big|_{t=0} = \lim_{t \rightarrow 0} \frac{1}{t} (\pi(\exp(tX))f - f)$$

is defined, and if, moreover, $g \mapsto \pi(g)Xf$ is a continuous function of G .

- We recursively define f to be C^k , $k \geq 1$ if f is C^1 and Xf is C^{k-1} for all $X \in \mathfrak{g}$.

- We say f is C^∞ or smooth if it is C^k for all $k \geq 1$.

Lemma 2.4.1: The space H^∞ is invariant under the action of G . Proof: pg 198

Proposition 2.4.1: Let (π, H) be a Hilbert space representation of the group $G = GL(n, \mathbb{R})$ or $G = GL(n, \mathbb{R})^+$. Then the action of $\mathfrak{g}$ on the space H^∞ of smooth vectors is a Lie algebra representation. Proof: see next section? Technical: m

Proposition 2.4.2: Let (π, H) be a Hilbert space representation of either $G = GL(n, \mathbb{R})$ or $G = GL(n, \mathbb{R})^+$.

(i) If $\phi \in C_c^\infty(G)$ (compactly supported, smooth), $f \in H$, then $\pi(\phi)f \in H^\infty$.

Here, $\pi(\phi): H \rightarrow H$ given by $\pi(\phi)f = \int_G \phi(g) \pi(g)f dg$, (averaging)

(ii) the space H^∞ is dense in H .

Proof: intuitive idea: mollifiers make smooth
choose approximation of identity.

(4)

Assume $G = GL(n, \mathbb{R})$, $K = O(n)$ or $G = GL(n, \mathbb{R})^+$, $K = SO(n)$.

def: A representation (π, H) of G on a Hilbert space H is called admissible if each (isomorphism class of finite-dimensional unitary representation of K occurs only finitely many times in some decomposition of H into irreducible unitary representations of K .

Unproven Fact: Any irreducible unitary representation is admissible.

Goal: Prove that in any irreducible admissible unitary representation of $G = GL(2, \mathbb{R})^+$ the multiplicity of any irreducible K -representation is at most one. ($K = SO(2)$).

use $\hookrightarrow$ Theorem 2.4.3: Let $G = GL(2, \mathbb{R})^+$ and $K = SO(2)$, and let $k \in \mathbb{Z}$. Let (π, H) be an irreducible unitary representation of G . Let H_k be the space of vectors $H_k = \{v \in H \mid \pi(K_\theta)v = \sigma_k(K_\theta)v\}$, where $\sigma_k(K_\theta) = e^{ik\theta}$. Assume that for $\phi \in C_c^\infty(K \backslash G/K, \sigma_k)$, the operator $\pi(\phi)$ is a compact operator on H_k . Then $\dim H_k \leq 1$.

use $\hookrightarrow$ Proposition 2.7.8: Let $G = GL(2, \mathbb{R})^+$ and let $K = SO(2)$. Let σ be a character of K . Then the algebra $C_c^\infty(K \backslash G/K, \sigma)$ is commutative.

$\hookrightarrow$ Theorem 2.4.2: (Uniqueness of K -fixed vector):
Let $G = GL(n, \mathbb{R})$ and $K = O(n)$, or else $G = GL(n, \mathbb{R})^+$ and $K = SO(n)$. Let (π, H) be an irreducible unitary representation of G . Assume that for $\phi \in C_c^\infty(K \backslash G/K)$, the operator $\pi(\phi)$ is a compact operator on H^K .
• $H^K = \{v \in H \mid \pi(k)v = v \ \forall k \in K\}$ closed. $\pi(\phi)$ preserves H^K .
Then $\dim H^K \leq 1$.

• If $\phi_1, \phi_2 \in C_c^\infty(G)$, where $G = GL(n, \mathbb{R})$, $GL(n, \mathbb{R})^+$, the convolution $\phi_1 * \phi_2 \in C_c^\infty(G)$ is defined by $(\phi_1 * \phi_2)(g) = \int_G \phi_1(gh) \phi_2(h^{-1}) dh = \int_G \phi_1(h) \phi_2(h^{-1}g) dh$.

$\hookrightarrow$ This makes $C_c^\infty(G)$ a ring (without unit). Noncommutative!

• Denote by $C_c^\infty(K \backslash G/K)$ the subring of functions

$$C_c^\infty(K \backslash G/K) := \{\phi \in C_c^\infty(G) \mid \phi(k_1 g k_2) = \phi(g) \text{ for } k_1, k_2 \in K\}.$$

$\hookrightarrow$ analogous to a "Hecke algebra". Commutative!

Theorem 2.2.3 (Gelfand): Let $G = GL(n, \mathbb{R})$, $K = O(n)$, or else let $G = GL(n, \mathbb{R})^+$, $K = SO(n)$. Then $C_c^\infty(K \backslash G/K)$ is commutative.

Proof idea: Transposition is an involution. $K \backslash G/K$ has symmetric representatives $\Rightarrow$ anti-involution is identity! $\square$ (ps 163)

$\hookrightarrow$ Proof of Prop 2.2.8: use $g \leftrightarrow (-,)^* g (-,)$ instead of transpose. $\square$

• If σ is a character of K , a homomorphism $\sigma: K \rightarrow \mathbb{C}^\times$, define $C_c^\infty(K \backslash G/K, \sigma)$ to be $C_c^\infty(K \backslash G/K, \sigma) := \{\phi \in C_c^\infty(G) \mid \phi(k_1 g k_2) = \sigma(k_1) \phi(g) \sigma(k_2) \text{ for } k_1, k_2 \in K\}$.

$$K_0 = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

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Def: An operator $T: H \rightarrow H$ is compact or completely continuous if T maps bounded sets into compact sets. (pg 166, Spectral Thm).

Fact: Any operator on a finite dimensional vector space is compact.

Proof of Thm 2.4.2 (Uniqueness of K -fixed vector):

① First Goal: Show H^K (if nonzero) is irreducible as a $C_c^\infty(K \backslash G/K)$ ^{-module}

$\hookrightarrow$ i.e., the only closed subspaces of H^K invariant under $\pi(\phi)$ for $\phi \in C_c^\infty(K \backslash G/K)$ are H^K and $\{0\}$.

② Second Goal: Prove H^K is finite dimensional.

Finish: As $C_c^\infty(K \backslash G/K)$ is a commutative family of normal operators on finite dimensional vector space, can simultaneously diagonalize. $\Rightarrow \dim 1$. ☺

Proof of ①: Suppose $V_0 \subset H^K$ is a closed nonzero subspace invariant under $C_c^\infty(K \backslash G/K)$

If $V_0 \neq H^K$, then $\exists x \in H^K \setminus \{0\}$ orthogonal to V_0 .

• Let $X = \{\pi(\phi)x \mid \phi \in C_c^\infty(G)\} \Rightarrow X \neq \{0\}$

$\hookrightarrow$ Prove that X is orthogonal to V_0 . (pg 194-195) $\Rightarrow X \perp H^K$

• "It is not hard to see X is G -invariant."

$\hookrightarrow$ Contradict irreducibility of π ! $\times$

Proof of ②: WTS H^K is finite dim, knowing it's irreducible. Assume nonzero.

Let $x \in H^K \setminus \{0\}$.

use $\hookrightarrow$ Lemma 2.3.2: Let $\pi: G \rightarrow \text{End}(H)$ be a unitary rep of G on a Hilb. H with $\xi \neq 0$. Let $0 \neq f \in H$. Let $\varepsilon > 0$ given. Then $\exists \phi \in C_c^\infty(G)$ s.t. $\pi(\phi)$ is

self-adjoint and $\|\pi(\phi)f - f\| < \varepsilon$. (suppress)

(In particular, if $\varepsilon \leq \|f\|$, this implies $\pi(\phi)f \neq 0$.)

Moreover if $\pi(k_0)f = e^{ik_0\theta}f$ for all $k_0 \in K = S^1 \cap \mathbb{Z}$, then we may choose ϕ so that $\phi(k_0g) = \phi(gk_0) = e^{ik_0\theta}\phi(g)$.

Furthermore if the space $H(k) := \{v \in H \mid \pi(k_0)v = e^{ik_0\theta}v\}$ is finite dim then we find ϕ' such that $\pi(\phi')f = f$.

• By Lemma 2.3.2,

$\exists \phi \in C_c^\infty(G)$ such that $\pi(\phi)$ is selfadjoint, $\pi(\phi)x \neq 0$, and (taking $k=0$ in the lemma), $\phi \in C_c^\infty(K \backslash G/K)$. ($f \in H^K$)

• By Hypothesis, $\pi(\phi)$ is compact.

$\hookrightarrow$ Apply Spectral Thm! $\Rightarrow \pi(\phi)$ has nonzero eigenvalue λ on H^K

• Let $L \subset H^K$ be λ -eigenspace of $\pi(\phi)$.

$\hookrightarrow L$ is nonzero, $\dim L < \infty$. (spectral thm).

• Because H^K is irreducible, $H^K = L$. $\square$

pg 2.7.8

Proof of Thm 2.4.3: Same argument, but use $C_c^\infty(K \backslash G/K, \sigma)$ commutative, instead. $\square$

$\hookrightarrow$ Conclude Goal: H^K has $\dim \leq 1$ b.c. admissibility $\Rightarrow \dim H_K < \infty \Rightarrow \pi(\phi)$ compact on H_K . $\square$

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Next concept: K-finite vectors

- Let (π, H) be an admissible representation of $G = GL(n, \mathbb{R})$ or $GL(n, \mathbb{R})^+$.
 $\rightarrow$ WLOG $\pi|_K$ is unitary.
- Write $H(\sigma)$ to denote σ -isotypic part of H . (σ is an irred. unit. rep of K)
 $\hookrightarrow H(\sigma) = \text{sum of all irred. invariant subspaces of } H \text{ that are isomorphic to } \sigma$
- Exercise 2.4.1: $H = \bigoplus H(\sigma)$ (Hilbert space direct sum).
- Admissibility means $\dim H(\sigma) < \infty$ for all σ in unitary dual of K .

Def: Write $H_{\text{fin}} := \bigoplus H(\sigma)$ (algebraic direct sum).

- i.e. H_{fin} is the space of all vectors $f \in H$ whose orthogonal projection on only a finite number of $H(\sigma)$ is nonzero.
- Elements of H_{fin} are called K-finite vectors.

Notation: Write $\mathfrak{h}_0 = \text{Lie}(K)$. Exercise (2.2.6(a)): $\mathfrak{h}_0 = \mathfrak{so}(2) = \{X \in M_2(\mathbb{R}) \mid X + X^t = 0\}$

Proposition 2.4.4: In this situation, the following are equivalent for $f \in H_{\text{fin}}^{\infty}$ (smooth).

- (i) f is a K-finite vector
- (ii) The vector space $\text{span} \{ \pi(k)f \mid k \in K \}$ is finite dimensional
- (iii) The vector space $\text{span} \{ Xf \mid X \in \mathfrak{h}_0 \}$ is finite dimensional.

Proof: kinda obvious? see p. 197. $\square$

Proposition 2.4.5 Let (π, H) be an admissible Hilbert space representation of $G = GL(n, \mathbb{R})$ or $GL(n, \mathbb{R})^+$. The K-finite vectors are smooth.

The space H_{fin} is dense in H and is invariant under the action of $\mathfrak{g} = \mathfrak{gl}(n, \mathbb{R})$ on H_{fin}^{∞} .

Proof: Technical. A more modern proof see [10].

Next concept: $(\mathfrak{g}, K)$ -module

Def: Let G be a reductive group, K a maximal compact subgroup of G , $\mathfrak{g}_0 = \text{Lie}(G)$.
 (Ex: $G = GL(n, \mathbb{R})^+$, $K = SO(n)$, $\mathfrak{g}_0 = \mathfrak{gl}(n, \mathbb{R})$ or $\mathfrak{gl}(n, \mathbb{R})$, $K = O(n)$, $\mathfrak{g}_0 = \mathfrak{gl}(n, \mathbb{R})$).

Def: A $(\mathfrak{g}, K)$ -module is a vector space V together with representations π of $\mathfrak{g}$ and σ of K (also denoted π), such that

- (i) V decomposes into an algebraic direct sum of finite dimensional invariant subspaces under K
- (ii) Representations of $\mathfrak{g}$ & K compatible: $\pi(X)f = Xf = \frac{d}{dt} \pi(\exp(tX)f) \big|_{t=0}$ (action)
- (iii) $\pi(g)\pi(X)\pi(g^{-1})f = \pi(\text{Ad}(g)X)f$ valid when $X \in \mathfrak{h}$ and $f \in V$.